

Sampling variability

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Confounders and controls

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Baselines and fair comparisons

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Replication and reproduction

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Sampling variability. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Sampling variability. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Population 0, 0, 4, 4 with distinct IDs; enumerate size 2 sample means and their average.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Find the mean of sample means and the chance of mean 2 under uniform selection of those six samples. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Population0, 0, 4, 4 with distinct IDs; enumerate size2 sample means and their average.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled sampling variability evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of sampling variability. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

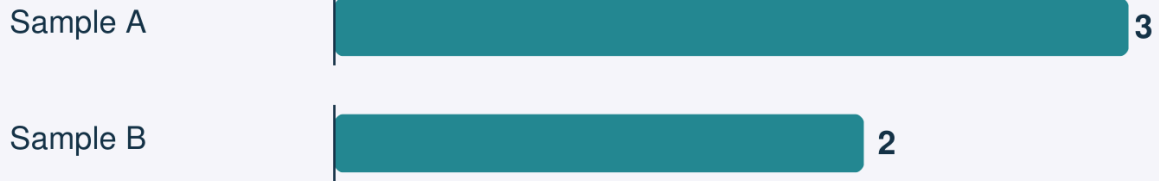
12. Pictorial case practice: Sampling variability

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

Sample A has 3 correct of 4; sample B has 2 correct of 4. The model and rubric are unchanged.

Two distinct samples, fixed model

correct cases out of 4 in each sample



Bar length shows the supplied value; the marked baseline is zero.

Sampling variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep model fixed

No training or threshold change occurs between samples.



2. Compute observed rates

A is 75%; B is 50%, differing by 25 percentage points.



3. Consider sampling

Different included cases can change a small-sample rate.

CHANGE ONE THING

Evaluate the exact same fixed four cases twice with deterministic predictions.

Explain the changed result using the matching lesson.

Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Different included cases can change a small-sample rate.

1. Consider sampling
2. Keep model fixed
3. Compute observed rates

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Consider sampling Different included cases can change a small-sample rate. Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

Practice 2: explain the deciding evidence

Changed-case practice: Evaluate the exact same fixed four cases twice with deterministic predictions. Which conclusion is supported by the supplied facts?

1. The counts should agree unless some condition, data or scoring changed.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Repair a missing building block

Review Confounders and controls (ai-7/10). Goal: Explain the mechanism and terminology of confounders and controls. Short practice: Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the mechanism and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the mechanism and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
